

Does the Interlingua Grok? Tracking the Emergence of Cross-Lingual Alignment via Mechanistic Interpretability Progress Measures

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Draft: March 24, 2026

1. Introduction and Motivation

Multilingual language models like mBERT, XLM-R, and NLLB achieve remarkable cross-lingual transfer — a classifier trained on English representations generalizes to French, Chinese, or Arabic without additional training. The standard explanation invokes a shared representation space: semantically equivalent inputs in different languages converge to similar internal representations, enabling task heads trained in one language to generalize. But this explanation is static. It describes the endpoint, not the process. How does cross-lingual alignment emerge during training? Is it gradual or sudden? Does it look like the memorization-then-generalization dynamic observed in grokking?

Nanda et al. (2023) fully reverse-engineered a one-layer transformer trained on modular addition and discovered that grokking — the phenomenon where a model abruptly generalizes long after memorizing its training data — arises from three continuous phases: memorization, circuit formation, and cleanup. Critically, the generalizing algorithm (a Fourier multiplication circuit) develops gradually in the weights well before the phase transition becomes visible in test accuracy. By defining continuous progress measures that track the underlying algorithm, Nanda showed that what appears to be sudden emergence is actually the late-stage reveal of a structure that was building all along.

We propose to apply this framing to multilingual language models. Our central question: does cross-lingual alignment emerge through a grokking-like dynamic, where language-specific memorization dominates early training while a shared cross-lingual circuit forms gradually in the background, becoming visible only when capacity pressure (analogous to weight decay in grokking) suppresses the language-specific components?

Understanding this process is not only scientifically interesting — it is practically urgent. Models like NLLB support 200 languages but performance remains uneven, with low-resource languages lagging significantly. If we understand the phase structure of cross-lingual alignment, we can diagnose *why* specific low-resource languages fail to align and design targeted interventions — optimal curricula, bridge languages, feature-level bootstrapping — rather than relying on the blunt instrument of more data and compute.

This proposal describes a two-phase research program. **Phase 1** establishes the mechanistic framework: defining progress measures, characterizing the phase structure of cross-lingual alignment, and testing the grokking analogy across training objectives, regularization regimes, and curricula. **Phase 2** applies this framework to low-resource translation: diagnosing failure modes for specific low-resource languages and designing mechanistically-informed interventions to close the performance gap.

This work sits at the intersection of three active research fronts — multilingual representation learning, mechanistic interpretability, and low-resource NLP — and addresses a gap that none has filled alone.

2. Related Work

2.1 Grokking and Phase Transitions in Training Dynamics

Power et al. (2022) first observed grokking in small transformers on algorithmic tasks. Nanda et al. (2023) provided a mechanistic explanation via reverse engineering, defining progress measures that tracked circuit formation before the behavioral phase transition. Liu et al. (2022) constructed phase diagrams with four phases (comprehension, grokking, memorization, confusion), showing that grokking is sensitive to hyperparameters, particularly weight decay. Subsequent work has found grokking in more complex settings involving images, text, and tabular data (Thilak et al., 2022; Gromov, 2023).

2.2 Training Dynamics of Multilingual Models

Wang, Minervini & Ponti (ACL Findings 2024) probed the emergence of cross-lingual alignment during BLOOM's training using neuron overlap analysis. They found high correlation between cross-lingual neuron overlap and downstream transfer performance, but also detected non-monotonic dynamics — degradation of alignment in certain training phases followed by recovery. This is suggestive of a more complex phase structure than simple gradual improvement, but they did not characterize it in terms of grokking or phase transitions.

Blevins et al. (EMNLP 2022) trained an XLM-R replica from scratch with checkpoints and probed for linguistic properties throughout. They found that in-language and cross-language abilities emerge at different rates, with cross-lingual transfer lagging behind monolingual

performance — consistent with a two-phase dynamic where language-specific learning precedes cross-lingual generalization.

"When Meanings Meet" (Dumas et al., January 2026) applied cross-lingual concept patching across 26 EuroLLM checkpoints — the finest-grained checkpoint analysis to date for a multilingual LLM. They found that alignment with shared concept spaces depends on training data proportion, and that different training phases (EuroLLM's two-phase curriculum) produce qualitatively different alignment patterns. This is the strongest existing evidence for phase-dependent dynamics, but it does not use mechanistic interpretability tools (SAEs, circuits) or define formal progress measures.

"Revisiting Multilingual Data Mixtures" (October 2025) ran controlled curriculum experiments on 3B-parameter models, comparing all-at-once training, English-first, and pivot-languages-first curricula. They found that curriculum order affects learning dynamics — including forgetting effects at transition points — but converges to similar final multilingual performance. This is directly relevant to our Experiment B: they established that the endpoint is stable across curricula, but measured only loss curves and downstream accuracy, not mechanistic progress measures. Our proposal extends their experimental design with the interpretability tools needed to characterize *why* different curricula produce different trajectories.

"Crosscoding Through Time" (September 2025) is the most directly relevant prior work. They deployed crosscoders — SAE variants that learn joint feature spaces across model checkpoints — on BLOOM, and found that monolingual models transition from token-specific detectors to high-level syntactic patterns, while multilingual models consolidate these into universal cross-lingual features over training. This is evidence for the gradual formation of shared features, tracked at the SAE level. However, they did not connect this to grokking dynamics, did not define progress measures, and did not examine whether the behavioral transition (downstream transfer accuracy) lags behind the feature-level transition.

2.3 Mechanistic Interpretability of Multilingual Models

Chi et al. (ACL 2020) used structural probes to find universal grammatical relations in mBERT — cross-lingual syntactic subspaces and emergent dependency label clusters. This established that something like a structural interlingua exists in the final model, but provides no information about how or when it forms during training.

"Tracing Multilingual Representations with Cross-Layer Transcoders" (November 2025) used Circuit Tracing methods (CLTs and attribution graphs) on purpose-trained multilingual models and found shared features across languages with language-specific decoding in late layers. Training without English produced identical shared-space structures. This establishes the endpoint but not the trajectory.

Anthropic's Circuit Tracing work (Ameisen, Lindsey et al., March 2025; Lindsey, Gurnee et al., March 2025) found evidence that Claude 3.5 Haiku has a shared conceptual space where reasoning occurs before being translated into output language. This is mechanistic

evidence for an interlingua at the circuit level, but again describes the converged model, not the formation process.

2.4 Abstraction vs. Exemplar Learning During Training

Jian & Manning (EACL 2026) provide the most direct behavioral evidence for phase-transition-like dynamics in linguistic category learning. Tracking GPT-2 across 450 training checkpoints, they find that linguistic categories are learned **abstraction-first**: class-level behavior (e.g., all dative verbs treated similarly) emerges *before* item-specific behavior (individual verb idiosyncrasies). Different linguistic constructions emerge abruptly and sequentially — syntactic subcategorization first ($t < 100$ steps), semantic verb argument classes next ($t > 100$), then non-local filler-gap dependencies later still. Before each onset, their divergence metrics show no change, then a sharp transition — a grokking-like pattern for linguistic abstractions, though they do not use that framing.

Critically, they validate their findings against an exemplar-first baseline (count-based co-occurrence vectors), confirming that the observed abstraction-first trajectories are a property of the transformer's learning dynamics, not of the training data distribution. Their JSD-based divergence metrics — which distinguish class-level from item-level learning — are cheap to compute (forward passes only, no SAE training) and transfer directly to the multilingual setting by replacing "verb class" with "language" and "item-specific" with "language-specific."

This work operates entirely at the behavioral level (next-token distributions) and is monolingual (English GPT-2). Our proposal extends it in two directions: mechanistically (using SAEs and circuits to characterize what internal shifts produce the behavioral transitions they observe) and multilingually (asking whether cross-lingual class-level behavior emerges before language-specific behavior).

2.5 The Gap

No existing work combines all of the following:

1. Training checkpoint analysis of multilingual models
2. Mechanistic interpretability tools (SAEs, circuits, attribution graphs) as progress measures
3. The grokking framing — testing whether cross-lingual alignment follows a memorization → circuit formation → cleanup dynamic
4. Formal continuous progress measures that precede the behavioral phase transition
5. Distributional divergence metrics (Jian & Manning, 2026) extended from monolingual category learning to cross-lingual abstraction

This proposal addresses that gap.

3. Research Questions and Hypotheses

RQ1: Does cross-lingual alignment emerge through a grokking-like dynamic?

H1 (Grokking hypothesis): During multilingual model training, cross-lingual alignment — as measured by mechanistic interpretability progress measures — increases continuously before the downstream cross-lingual transfer accuracy shows a sharp improvement. The behavioral transition (transfer accuracy) lags behind the mechanistic transition (shared features/circuits), analogous to grokking.

H1-alt (Gradual hypothesis): Cross-lingual alignment and downstream transfer accuracy increase at similar rates throughout training, with no lag between mechanistic and behavioral measures.

RQ2: Do the three grokking phases (memorization, circuit formation, cleanup) have multilingual analogs?

H2: Early training is dominated by language-specific memorization (language-specific SAE features, language-specific circuits). Shared cross-lingual features and circuits develop gradually during a "circuit formation" phase. The apparent onset of cross-lingual transfer corresponds to a "cleanup" phase where capacity pressure (shared parameters across languages, weight decay) suppresses language-specific components and reveals the shared structure.

RQ3: Does the training objective affect the phase structure?

H3a: Models without explicit cross-lingual objectives (mBERT, XLM-R) show a more pronounced grokking dynamic — the interlingua is "accidental" and emerges later. Models with explicit cross-lingual objectives (NLLB) show earlier and more gradual alignment.

H3b: Capacity pressure (number of languages sharing parameters) plays the role of weight decay in grokking. More languages → stronger pressure → faster transition from memorization to shared representations, paralleling the finding in grokking that stronger regularization accelerates the phase transition.

H3c: Curriculum-induced pressure onset produces the most dramatic grokking signature. A model trained first on English alone builds English-specific circuits with no sharing pressure. When additional languages are introduced into the same parameter space mid-training, the sudden increase in capacity pressure should trigger rapid circuit formation — analogous to increasing weight decay mid-training. The progress measures should show a sharp onset of cross-lingual feature emergence at the curriculum switch point, with downstream transfer accuracy lagging behind.

RQ4: Do typologically distant languages show different phase transition timing?

H4: Language pairs with greater typological similarity (e.g., English–French) show earlier cross-lingual alignment than typologically distant pairs (e.g., English–Chinese, English–Arabic), consistent with Karthikeyan et al.'s finding that structural similarity drives transfer. The progress measures should show this differential timing before it appears in downstream accuracy.

4. Progress Measures

We define three families of continuous progress measures, each tracking a different aspect of cross-lingual alignment. The key test is whether these measures increase smoothly while downstream cross-lingual transfer accuracy appears to jump.

4.1 Representation Geometry Measures

CKA similarity between language pairs. For each training checkpoint, compute Centered Kernel Alignment (Kornblith et al., 2019) between representations of translationally equivalent sentences in different languages, at each layer. Track CKA as a function of training step.

Mean-centered representation overlap. Following Adams et al. (2022), mean-center representations per language and measure the alignment of language-neutral axes (encoding token position, POS) across checkpoints.

4.2 Feature-Level Measures (SAE-based)

Cross-lingual feature activation fraction. Train SAEs on the residual stream at selected layers for each checkpoint. Measure the fraction of SAE features that activate on inputs from multiple languages. A feature that fires on both English and French syntactic subjects is evidence for a shared representation; one that fires only on English subjects is language-specific.

Feature splitting dynamics. Track whether features that are initially cross-lingual split into language-specific variants as dictionary size increases (connecting to Bricken's feature splitting phenomenon). Conversely, track whether initially language-specific features merge into cross-lingual features over training.

Crosscoder overlap. Following "Crosscoding Through Time," train crosscoders that learn joint feature spaces across language pairs at each checkpoint. Measure the fraction of features shared vs. language-specific as a function of training step.

4.3 Circuit-Level Measures

Circuit overlap for syntactic tasks. Using Edge Pruning (Bhaskar et al., 2024) or Attribution Patching (Syed et al., 2024), extract circuits for subject-verb agreement in each language at each checkpoint. Measure the Jaccard similarity of circuit edges across language pairs as a function of training step.

Attribution graph similarity. For models large enough to support Circuit Tracing (CLTs), compare attribution graphs for translationally equivalent prompts across languages. Measure feature-level overlap in the active attribution graph.

4.4 Distributional Divergence Measures (JSD-based)

Following Jian & Manning (2026), we define Jensen-Shannon divergence metrics over next-token prediction distributions that distinguish cross-lingual class-level learning from language-specific item-level learning. These metrics are computationally cheap (forward passes only, no SAE training required) and provide fine-grained behavioral tracking at every checkpoint.

Cross-lingual class divergence. For a syntactic construction (e.g., subject-verb agreement), collect next-token distributions at the target position across verbs in different languages. Compute pairwise JSD between distributions conditioned on the same verb class in different languages (cross-lingual within-class) vs. different verb classes in different languages (cross-lingual between-class). If cross-lingual within-class JSD is significantly lower than between-class JSD, the model has learned to treat the syntactic category similarly across languages — evidence for class-level cross-lingual abstraction.

Language-specific item divergence. Compute pairwise JSD between distributions for individual verbs within the same language and syntactic class. Increasing within-class JSD indicates item-specific learning (individual verb idiosyncrasies). The grokking prediction: cross-lingual class divergence should emerge before language-specific item divergence, paralleling Jian & Manning's finding that class-level behavior precedes item-level behavior in monolingual GPT-2.

Sequential onset detection. Following Jian & Manning's methodology, track the onset of learning for different linguistic constructions across languages. If constructions emerge sequentially (e.g., POS categories before argument structure before non-local dependencies) and the ordering is consistent across languages, this is evidence for shared abstract learning mechanisms — strengthening the case for a computational interlingua.

Exemplar-first baseline. Construct count-based co-occurrence baselines for each language (following Jian & Manning's methodology) to confirm that observed abstraction-first trajectories are a property of the model's learning dynamics, not the data distribution.

4.5 Behavioral Measures (for comparison)

Zero-shot cross-lingual transfer accuracy. Train a task head (NER, POS tagging, dependency parsing) on English at each checkpoint; evaluate on other languages without fine-tuning. This is the coarse behavioral measure that the progress measures should precede.

Structural probe accuracy on held-out languages. Following Chi et al. (2020), train structural probes on English dependency trees and evaluate cross-lingually at each checkpoint.

5. Experimental Design

5.1 Models

We propose a three-tier experimental design, each tier increasing in scale and ecological validity.

Tier 1: Controlled small-scale training (primary experiments). Train small multilingual transformers from scratch with frequent checkpointing (every 1000 steps). This follows the precedent of Blevins et al. (2022) and the "Tracing Multilingual Representations" paper, which found that training dynamics patterns replicate at small scale (TinyStories-sized models show the same multilingual organization as larger ones).

Base models (testing H1, H2, H3a):

- **Encoder-only (mBERT-like):** 6-layer, 512-dim, 8-head transformer with MLM objective on concatenated multilingual Wikipedia. No cross-lingual objective. This is the "minimal case" — any alignment is emergent.
- **Encoder-only (XLM-R-like):** Same architecture but with RoBERTa-style training (no NSP, more data, byte-pair encoding shared across languages). Tests whether removing the NSP confound (Karthikeyan et al.) changes the phase structure.
- **Encoder-decoder (NLLB-like):** Same-scale model with translation objective on parallel data. Tests H3a — whether explicit cross-lingual training changes the grokking dynamic.

Regularization note: There is an important asymmetry in how these models are regularized in practice. mBERT and XLM-R use AdamW with constant weight decay of 0.01 — the direct analog to the grokking setup, where weight decay provides persistent pressure toward compact (and thus shared) solutions. NLLB and the broader fairseq translation recipe typically use **no weight decay** (`--weight-decay 0.0`), relying instead on dropout, attention dropout, layerdrop, label smoothing, and for MoE variants, token dropout as regularization. These are fundamentally different mechanisms: weight decay continuously shrinks all weights toward zero (favoring parameter-efficient solutions), while dropout stochastically removes information during training (favoring redundant, noise-robust representations).

This means the H3a comparison between the mBERT-like and NLLB-like models is confounded by both the training objective (MLM vs. translation) and the regularization mechanism (weight decay vs. dropout). To disentangle these:

- Train an additional **mBERT-like variant with dropout but no weight decay**, matching the NLLB regularization approach while keeping the MLM objective. If this variant shows different phase dynamics than the standard mBERT-like model, the regularization mechanism matters independently of the training objective.
- Train an additional **NLLB-like variant with weight decay 0.01**, adding grokking-style pressure to the translation model. If weight decay accelerates cross-lingual alignment formation even in a model with an explicit cross-lingual objective, that strengthens the analogy.

Experiment A — Capacity pressure as regularization (testing H3b): Using the mBERT-like architecture, train three variants with identical total data but varying numbers of languages: 2 (English + French), 4 (adding Chinese, Arabic), and 8 (full set). Model capacity is held fixed. If capacity pressure plays the role of weight decay, more languages should accelerate the phase transition in the progress measures. This directly tests whether the number of languages sharing parameters is the multilingual analog of weight decay strength.

Experiment B — Curriculum-induced pressure onset (testing H3c): Using the mBERT-like architecture with all 8 languages, train four curriculum variants:

- **All-at-once (baseline):** All 8 languages from step 0.
- **English-first:** English only for the first 25% of training steps, then all 8 languages.
- **Pivot-first:** English + French + Chinese + Arabic for the first 25%, then all 8.
- **Incremental:** Add one language every 12.5% of training (English alone → +French → +Chinese → etc.).

The English-first condition should produce the most dramatic grokking signature: the model builds English-specific circuits during the monolingual phase, then faces sudden massive capacity pressure at the switch point when seven languages must share the same parameters. The progress measures should show rapid cross-lingual feature emergence at the switch, with downstream transfer lagging behind. This design follows the precedent of "Revisiting Multilingual Data Mixtures" (2025), which ran similar curriculum experiments but measured only final performance and loss curves — not mechanistic progress measures.

Language selection: 8 languages spanning 4 typological families — English, French (Indo-European, Germanic/Romance); Chinese, Japanese (Sino-Tibetan/Japonic); Arabic, Hebrew (Semitic); Finnish, Turkish (Uralic/Turkic). This supports testing H4 (differential phase transition timing by typological distance).

Checkpointing: Every 1000 optimization steps for base models and Experiment A, saving full model weights, optimizer state, and a fixed evaluation set of parallel sentences for all 8 languages. For Experiment B curriculum variants, increase checkpointing frequency to every

200 steps in a window of ± 5000 steps around each curriculum switch point, to capture rapid dynamics at the moment of pressure onset.

Tier 2: Analysis of existing checkpoints. Apply the same progress measures to publicly available checkpoint series:

- **BLOOM** (Wang et al., 2024 used 4–8 checkpoints; we would use all available intermediate checkpoints)
- **EuroLLM** (26 checkpoints, used by "When Meanings Meet")
- **OLMo-2** or **Apertus** (recently released with checkpoints)

This tier validates whether patterns found in Tier 1 replicate in larger, independently trained models.

Tier 3: Replication on focal course models. Apply progress measures to mBERT and XLM-R (final checkpoints only, since no training checkpoints are public) as a cross-sectional comparison. These are the course's focal models and connect directly to Chi et al.'s findings. Comparison with Tier 1 small-scale analogs tests whether the small models are faithful proxies.

5.2 Evaluation Tasks

For each checkpoint, evaluate:

- **POS tagging** (Universal Dependencies, 8 languages)
- **Dependency parsing** (UD, 8 languages) — enables Chi-style structural probes
- **Named entity recognition** (WikiANN, 8 languages)
- **Subject-verb agreement** (custom minimal pairs, 8 languages) — enables circuit extraction

All tasks use zero-shot cross-lingual transfer: train on English, evaluate on all 8 languages.

5.3 SAE Training Protocol

At selected checkpoints (every 5000 steps for Tier 1; all available for Tier 2), train SAEs on the residual stream at layers 1, 3, and 5 (for a 6-layer model) or equivalent fractions of deeper models. Use the standard Bricken et al. architecture with L1 sparsity penalty. Dictionary sizes: $4\times$ and $8\times$ the model dimension, enabling feature splitting analysis.

For crosscoder analysis, train crosscoders following the "Crosscoding Through Time" methodology, learning joint feature spaces across language pairs (English–French, English–Chinese, English–Arabic, English–Finnish) at each selected checkpoint.

5.4 Circuit Extraction Protocol

At selected checkpoints, extract circuits for subject-verb agreement using Edge Pruning (Bhaskar et al., 2024). Run separately for each language. Compare circuits via:

- **Edge Jaccard similarity** across language pairs
 - **Component overlap** (which attention heads / MLP layers appear in circuits for multiple languages)
 - **Exclusive components** (heads/MLPs that appear only in one language's circuit)
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6. Analysis Plan

6.1 Testing the Grokking Hypothesis (H1)

Plot all progress measures and behavioral measures as a function of training step. The grokking signature is: progress measures increase continuously while behavioral measures show a sharp transition. Formally, fit sigmoid functions to each measure's trajectory and compare the inflection points. If the mechanistic measures' inflection points precede the behavioral measures' inflection points by a statistically significant margin, H1 is supported.

The JSD-based distributional measures (Section 4.4) serve as a critical intermediate layer between the mechanistic measures (SAEs, circuits) and the coarse behavioral measures (downstream transfer accuracy). They can be computed at every checkpoint cheaply, providing high temporal resolution. The prediction is a three-layer ordering: mechanistic measures move first (SAE features, circuit overlap), then JSD class divergence shows cross-lingual abstraction forming, then downstream accuracy jumps. If JSD class divergence and SAE cross-lingual feature fraction move in lockstep, that is convergent evidence from independent measurement approaches that the same underlying shift is being detected.

6.2 Phase Identification (H2)

Following Nanda et al.'s three-phase characterization:

- **Memorization phase:** Language-specific SAE features dominate; cross-lingual feature fraction is low; circuits for the same task in different languages share few edges; JSD class divergence across languages is near zero (no cross-lingual abstraction yet); training loss decreases but cross-lingual transfer accuracy remains at chance.
- **Circuit formation phase:** Cross-lingual features begin to emerge; circuit overlap between language pairs increases; JSD class divergence begins to rise (cross-lingual abstractions forming); progress measures rise; but behavioral transfer accuracy has not yet jumped.
- **Cleanup phase:** Language-specific features are suppressed (their activation norms decrease); cross-lingual features become dominant; JSD class divergence reaches

plateau; behavioral transfer accuracy jumps. Analogous to weight decay removing memorization components in grokking.

Test for three-phase structure by fitting piecewise linear models to the progress measure trajectories and identifying transition points.

6.3 Capacity Pressure and Curriculum Effects (H3)

Training objective comparison (H3a). Compare phase transition timing across the three base Tier 1 models (mBERT-like, XLM-R-like, NLLB-like). If H3a is correct, the NLLB-like model shows earlier and more gradual alignment. If the grokking dynamic is a general phenomenon of multilingual training, all three should show the same qualitative pattern but with different timing. Because the mBERT-like and NLLB-like models differ in both training objective and regularization mechanism (weight decay vs. dropout), the two control variants (mBERT-like without weight decay; NLLB-like with weight decay) are essential for attribution. If the mBERT-like model without weight decay shows slower or absent grokking dynamics while the NLLB-like model with weight decay shows faster alignment, that would establish weight decay — not the training objective — as the critical factor for the grokking-like phase structure.

Experiment A — Capacity pressure (H3b). Compare progress measure trajectories across the 2-language, 4-language, and 8-language mBERT-like variants. The key test: does the inflection point in the progress measures shift earlier as the number of languages increases? Plot phase transition timing as a function of number of languages. If the relationship is monotonic (more languages → earlier transition), this supports the capacity-pressure-as-regularization analogy. This parallels Liu et al.'s (2022) phase diagrams, which showed that stronger weight decay shifts the grokking transition earlier.

Experiment B — Curriculum (H3c). For each curriculum variant, align progress measures to the curriculum switch point (the step at which new languages are introduced). The English-first condition is the critical test: does the switch point trigger a rapid onset of cross-lingual features in the progress measures? Compare the slope of the progress measures immediately after the switch to the all-at-once baseline's slope at the same training step. A steeper slope after the switch is the grokking signature — sudden capacity pressure catalyzing circuit formation. Also compare to "Revisiting Multilingual Data Mixtures" (2025): they found that curriculum order affects dynamics but not final performance. Our progress measures should reveal *why* — different curricula may reach the same endpoint through different phase structures.

6.4 Typological Distance Effects (H4)

For each language pair, record the training step at which cross-lingual alignment (by each progress measure) crosses a threshold. Correlate threshold-crossing time with typological distance measures (lang2vec features, syntactic distance from WALS). If H4 is correct, closer languages align earlier.

Additionally, following Jian & Manning's finding that different linguistic constructions emerge sequentially in monolingual models, test whether the *ordering* of construction onsets is consistent across languages. If POS categories align cross-lingually before argument structure, which aligns before non-local dependencies — the same ordering Jian & Manning found monolingually — that is evidence for a universal abstraction hierarchy that transcends individual languages. If the ordering differs by language (e.g., morphologically rich languages show earlier cross-lingual alignment for agreement but later alignment for word order), that reveals how typological properties modulate the learning trajectory.

7. Phase 1 Expected Contributions

1. **A grokking-based framework for understanding cross-lingual emergence.** Connecting two previously separate literatures (training dynamics / phase transitions and multilingual representation learning) through the lens of mechanistic interpretability.
 2. **Continuous progress measures for cross-lingual alignment.** Generalizing Nanda et al.'s approach from a single circuit in a toy model to a complex phenomenon in realistic multilingual models. These include mechanistic measures (SAE feature overlap, circuit similarity) and distributional divergence measures extending Jian & Manning (2026) from monolingual category learning to cross-lingual abstraction. These measures could serve as early warning signals for when cross-lingual capabilities will emerge during training.
 3. **Feature-level and circuit-level characterization of the interlingua formation process.** Going beyond representation similarity (CKA) to ask whether the model builds shared computational pathways, and tracking when this happens.
 4. **Comparison across training objectives and curricula.** Establishing whether explicit cross-lingual training (NLLB-style) produces qualitatively different dynamics than emergent multilinguality (mBERT-style), and whether capacity pressure (number of languages) and curriculum design (timing of language introduction) modulate the phase structure. The curriculum experiments extend "Revisiting Multilingual Data Mixtures" by explaining *why* different curricula produce different training dynamics despite converging to similar endpoints.
 5. **Open-source checkpoints and analysis toolkit.** All Tier 1 model checkpoints, trained SAEs, extracted circuits, and progress measure computations will be released for reproducibility and extension.
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8. Phase 2: From Understanding to Intervention — Low-Resource Translation

Phase 1 establishes the diagnostic framework. Phase 2 applies it to the problem that motivates the entire enterprise: improving translation quality for low-resource languages.

8.1 Motivation

NLLB achieved 44% BLEU improvement over prior work across 200 languages, but performance remains deeply uneven. Low-resource languages — particularly those that are typologically distant from high-resource languages in the training set — lag significantly. Current approaches to closing this gap are blunt: mine more parallel data, increase model capacity, or oversample low-resource languages during training. These approaches are expensive and scale poorly. A mechanistic understanding of *why* specific languages fail to align enables targeted, efficient interventions.

8.2 Diagnostic Framework: Three Failure Modes

The grokking phase structure from Phase 1 provides a taxonomy of failure modes for low-resource languages, each diagnosable via the progress measures:

Failure Mode 1: Stuck in memorization. The model sees too few examples of the low-resource language to exit the memorization phase. The language's examples are stored as isolated lookup entries, never triggering the formation of shared cross-lingual circuits. Diagnostic signature: language-specific SAE features only, near-zero cross-lingual feature overlap, flat JSD class divergence — the language never enters phase two.

Failure Mode 2: Incomplete circuit formation. Shared representations begin to develop but the low-resource language's contribution to the gradient is too small relative to high-resource languages to sustain the process. The circuit formation signal is present but too weak to reach cleanup. Diagnostic signature: initial increase in cross-lingual features, then plateau or regression at a low level — progress measures stall mid-trajectory.

Failure Mode 3: Structural incompatibility. The low-resource language is typologically distant from all high-resource languages in the training set. The shared circuits that develop among high-resource languages encode structural regularities (e.g., SVO word order, prepositional phrases) that don't transfer to the low-resource language's structure (e.g., SOV, postpositional). Diagnostic signature: strong cross-lingual alignment among high-resource pairs, but the low-resource language's alignment with any partner remains flat across all progress measures.

Each failure mode suggests a different intervention. Applying the wrong intervention wastes resources: more data won't help Failure Mode 3, and a bridge language won't help Failure Mode 1. The diagnostic framework enables precise targeting.

8.3 Interventions

Intervention 1: Optimal curriculum design for low-resource inclusion.

Building on Experiment B from Phase 1, design curricula that specifically bootstrap low-resource languages:

- **Bridge language sequencing.** Introduce a low-resource language alongside a typologically similar high-resource language, rather than alongside all languages at once. For example, introduce Wolof alongside French (shared contact features, some structural overlap) rather than alongside all 200 languages simultaneously. The progress measures reveal exactly when alignment with the bridge language begins, providing a real-time signal of whether the strategy is working — far faster feedback than waiting for downstream BLEU scores.
- **Scaffolded introduction.** Train first on a small set of high-resource languages until the shared circuits are well-established (detected by progress measures reaching plateau), then introduce low-resource languages into that pre-structured parameter space. The prediction: the pre-existing shared circuits act as scaffolding — the low-resource language can slot into circuits that already exist rather than building them from scratch. Test by comparing progress measure onset times for a low-resource language introduced into a pre-trained multilingual model vs. trained from scratch alongside all languages.
- **Critical data thresholds.** For each low-resource language, identify the minimum data quantity needed to cross the memorization → circuit formation threshold. This is the analog of the grokking finding that there is a critical data fraction below which generalization never occurs. If you can identify this threshold for each language, you know exactly how much data mining effort is worthwhile — and when diminishing returns set in.

Intervention 2: Feature-level bootstrapping.

The SAE analysis from Phase 1 reveals *which specific features* are shared across high-resource languages. This enables targeted interventions for low-resource languages:

- **Missing feature diagnosis.** For a low-resource language that fails to align, identify which cross-lingual SAE features fire for high-resource languages but not for the low-resource language. For example, if a "subject-verb agreement" feature activates for English, French, and Arabic but not for Wolof, the model is not routing Wolof syntax through the shared circuit. This points to exactly what the model is failing to learn.
- **Targeted data design.** Rather than mining more general parallel data (expensive and often unavailable for low-resource languages), design or generate training examples that specifically exercise the missing features. If the model lacks a cross-lingual argument structure feature for Wolof, create examples that exercise dative constructions, motion verbs, and reciprocals in Wolof — the specific constructions that the shared circuits

handle for other languages. This is the linguistic analog of curriculum learning: present the model with examples designed to activate the generalizing circuit, not just more data.

- **Activation steering for translation.** The "Tracing Multilingual Representations" paper showed that language-specific decoding relies on a small set of high-frequency features in late layers, and intervening on these features allows suppression of one language and substitution of another. For low-resource translation, if the model's internal representation of a low-resource sentence has reached the shared conceptual space (detectable via SAE features) but the decoding pathway to the target language is weak, steering the decoding features may improve output quality without additional training data.

Intervention 3: Training-time monitoring and resource allocation.

Training large multilingual models is expensive. The progress measures provide principled answers to resource allocation questions:

- **When to stop training.** If cross-lingual feature overlap for a low-resource language has plateaued, more training won't help — a different intervention is needed (more data, curriculum change, bridge language).
- **When to reallocate compute.** If some low-resource languages are still in the circuit formation phase (progress measures rising) while others have stalled, dynamically shift training data sampling toward the languages that are still benefiting.
- **When to change strategy.** If progress measures diagnose Failure Mode 3 (structural incompatibility), no amount of the current training regime will help. This triggers a pivot to bridge language approaches or architectural modifications (e.g., language-specific adapter layers for structurally divergent languages).

8.4 Phase 2 Experimental Design

Target languages. Select 6–8 low-resource languages from the NLLB-200 set that span the three failure modes, guided by NLLB's published per-language BLEU scores and typological features:

- Languages with very low data volume and low BLEU (candidate Failure Mode 1)
- Languages with moderate data but low BLEU despite typological similarity to high-resource languages (candidate Failure Mode 2)
- Languages with moderate data but low BLEU and high typological distance from all high-resource languages (candidate Failure Mode 3)

Diagnostic phase. Apply the full progress measure suite from Phase 1 to NLLB-scale small models (or BLOOM checkpoints if available) for the target languages. Classify each language's failure mode based on diagnostic signatures.

Intervention experiments. For each failure mode, test the corresponding intervention:

- Failure Mode 1 languages: critical data threshold experiments + targeted data generation
- Failure Mode 2 languages: bridge language curricula + scaffolded introduction
- Failure Mode 3 languages: bridge language selection guided by SAE feature overlap + activation steering

Evaluation. FLORES-200 BLEU/chrF++ for the target low-resource languages, compared against baselines: (a) standard all-at-once training, (b) uniform oversampling of low-resource languages, (c) temperature-based sampling (the NLLB default). The key comparison is whether mechanistically-informed interventions outperform the standard approaches, and at what data/compute cost.

8.5 Phase 2 Expected Contributions

6. **A diagnostic taxonomy for low-resource translation failure.** Three mechanistically-grounded failure modes, each identifiable via progress measures, each suggesting a different intervention. This replaces the current one-size-fits-all approach (more data, more compute) with targeted diagnosis.
7. **Mechanistically-informed curriculum design for low-resource languages.** Bridge language sequencing, scaffolded introduction, and critical data thresholds — curriculum strategies derived from understanding the phase structure of cross-lingual alignment, validated against standard baselines on FLORES-200.
8. **Feature-level diagnosis and intervention for low-resource translation.** Using SAE features to identify exactly what a model is failing to learn for a specific low-resource language, and designing targeted training data or activation steering to address the specific gap.
9. **Progress measures as training-time monitoring tools.** Demonstrating that the mechanistic progress measures from Phase 1 can serve as practical real-time signals for resource allocation decisions during multilingual model training — a contribution to training methodology, not just scientific understanding.

9. Risks and Mitigations

Phase 1 Risks

Risk: Small-scale models may not exhibit grokking dynamics. Grokking is sensitive to hyperparameters (particularly weight decay and data fraction). Mitigation: We will sweep over weight decay values and training data fractions, following Liu et al.'s (2022) phase diagram

methodology. The "Tracing Multilingual Representations" paper provides evidence that multilingual dynamics replicate at small scale.

Risk: SAE training at every checkpoint is computationally expensive. Mitigation: Train SAEs at selected checkpoints (every 5000 steps) rather than every checkpoint. Use the crosscoder approach from "Crosscoding Through Time" which enables comparison across checkpoints without training independent SAEs at each one.

Risk: The grokking analogy may not hold — cross-lingual alignment may be genuinely gradual. This would support H1-alt, which is itself a contribution: establishing that the interlingua does not form through a phase transition would constrain theories of multilingual representation learning.

Risk: Experiment B's curriculum switch may not produce a grokking signature. The "Revisiting Multilingual Data Mixtures" paper found forgetting effects at curriculum transitions, which could mask the grokking signal. Mitigation: The high-frequency checkpointing around switch points (every 200 steps) is designed to capture rapid dynamics. Even if the behavioral signature is muddled by forgetting, the mechanistic progress measures may still show the onset of cross-lingual circuit formation beneath the surface — which would be the strongest version of the grokking finding.

Risk: Circuit extraction at scale is computationally demanding. Mitigation: Use Edge Attribution Patching (fast, approximate) for broad coverage across checkpoints and Edge Pruning (slower, more faithful) at key transition points identified by the progress measures.

Phase 2 Risks

Risk: Failure modes may not be cleanly separable. A low-resource language may exhibit a mixture of failure modes — some constructions stuck in memorization, others incompatible. Mitigation: The progress measures are computed per-construction and per-language-pair, allowing fine-grained diagnosis rather than a single label per language.

Risk: Interventions that work at small scale may not transfer to NLLB-scale models. Mitigation: Phase 2 experiments are designed at two scales — the Tier 1 small models (where controlled experiments are feasible) and existing NLLB/BLOOM checkpoints (where ecological validity is tested). Convergent findings across scales increase confidence.

Risk: Targeted data design requires linguistic expertise for each low-resource language. Mitigation: The SAE-based diagnosis identifies *which features* are missing, not what training examples to create. The translation from "missing subject-verb agreement feature" to actual training examples can leverage existing UD treebanks (which cover many low-resource languages) and LLM-assisted example generation, reducing the per-language linguistic expertise required.

Risk: Activation steering may introduce artifacts or degrade quality on non-targeted translation directions. Mitigation: Evaluate steering interventions on both the targeted low-resource direction and a held-out set of high-resource directions to ensure no regression.

10. Timeline

Phase 1: Mechanistic Framework (Summer)

Phase	Period	Activities
Setup	Summer Month 1	Data pipeline (multilingual Wikipedia, UD treebanks, parallel data); model architecture implementation; SAE and circuit extraction infrastructure
Tier 1 Base + Exp A	Summer Months 1–2	Train three base models + two regularization controls + three Experiment A variants with frequent checkpointing; begin SAE training on early checkpoints
Tier 1 Exp B	Summer Month 2	Train four Experiment B curriculum variants; checkpoint at high frequency around curriculum switch points
Tier 1 Analysis	Summer Months 2–3	Compute all progress measures (including JSD metrics) across all Tier 1 variants; extract circuits at selected checkpoints; test H1, H2, H3
Tier 2 Analysis	Summer Month 3	Apply progress measures to BLOOM and EuroLLM checkpoints; test replication
Tier 3 + Synthesis	Summer Month 4	Cross-sectional analysis on mBERT/XLM-R; compare across tiers; test H4; Phase 1 write-up

Phase 2: Low-Resource Translation Interventions (Fall/Winter)

Phase	Period	Activities
Language Selection	Fall Month 1	Select 6–8 target low-resource languages from NLLB-200 based on BLEU scores and typological profiles; acquire UD treebanks and parallel data for targets
Diagnostic	Fall Months 1–2	Apply full progress measure suite to NLLB-scale small models for target languages; classify each language's failure mode
Intervention Design	Fall Month 2	Design language-specific interventions based on diagnosed failure modes: bridge language curricula, targeted data, activation steering
Intervention Experiments	Fall Months 2–3	Train intervention models; compare against standard baselines on FLORES-200
Evaluation + Write-up	Fall Months 3–4	Full evaluation on target languages; regression testing on high-resource languages; synthesize Phase 1 + Phase 2 findings; paper preparation
Extension	Winter/Spring	Scale interventions to additional languages; explore integration with production training pipelines; follow-up publications

11. Required Resources

Phase 1

- **Compute:** GPU cluster access for training Tier 1 models. Base models: 3 models + 2 regularization controls × ~2 weeks each on 4× A100. Experiment A: 3 additional variants (2/4/8 languages) × ~2 weeks each. Experiment B: 4 curriculum variants × ~2 weeks each. Total Tier 1 training: ~12 model runs. SAE training: estimated 1× A100 per checkpoint, ~50 checkpoints per model at selected intervals. Circuit extraction via Edge Pruning at key transition points identified by progress measures.
- **Data:** Multilingual Wikipedia (public), Universal Dependencies treebanks (public), parallel data from OPUS for NLLB-like training (public).
- **Software:** TransformerLens, SAE training libraries (e.g., SAELens), Edge Pruning codebase (public, Princeton NLP), circuit-tracer library (public, Anthropic).

Phase 2 (additional)

- **Compute:** Intervention model training: ~6–8 models (one per target language × intervention type) at Tier 1 scale, plus diagnostic runs on BLOOM/EuroLLM checkpoints. Activation steering experiments require inference-time compute only (no training). Estimated additional ~8 model runs on 4× A100.
- **Data:** FLORES-200 evaluation set (public). UD treebanks for target low-resource languages (public, coverage varies). Parallel data for target languages from OPUS (public, quantity varies — data scarcity is part of what we're studying). Targeted training data may require generation via LLM-assisted methods or collaboration with native speaker linguists.
- **Software:** Phase 1 toolkit (reused). Additionally: FLORES-200 evaluation scripts, sacreBLEU/chrF++ evaluation, activation steering via circuit-tracer library.

Personnel (both phases)

- 1 PhD student or postdoc (full-time summer for Phase 1, continuing into fall for Phase 2)
- 1–2 MS students (supporting roles in data pipeline, SAE analysis, and evaluation infrastructure)
- For Phase 2 intervention design: access to linguistic expertise for target low-resource languages (via collaboration, UD treebank documentation, or native speaker consultants)
- PI oversight

References

Adams, O., Cotterell, R., & Gonen, H. (2022). The geometry of multilingual representations in XLM-R.

Ameisen, E., Lindsey, J., Pearce, A., et al. (2025). Circuit Tracing: Revealing Computational Graphs in Language Models. Transformer Circuits Thread.

Bhaskar, A., Wettig, A., Friedman, D., & Chen, D. (2024). Finding Transformer Circuits with Edge Pruning. NeurIPS 2024 (Spotlight).

Blevins, T., Gonen, H., & Zettlemoyer, L. (2022). Analyzing the mono- and cross-lingual pretraining dynamics of multilingual language models. EMNLP 2022.

Bricken, T., Templeton, A., Batson, J., et al. (2023). Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. Transformer Circuits Thread.

Chi, E. A., Hewitt, J., & Manning, C. D. (2020). Finding universal grammatical relations in multilingual BERT. ACL 2020.

Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S., & Garriga-Alonso, A. (2023). Towards Automated Circuit Discovery for Mechanistic Interpretability. NeurIPS 2023 (Spotlight).

Costa-jussà, M. R., Cross, J., Çelebi, O., et al. (NLLB Team). (2022). No Language Left Behind: Scaling Human-Centered Machine Translation. arXiv:2207.04672.

Crosscoding Through Time: Tracking Emergence & Consolidation Of Linguistic Representations Throughout LLM Pretraining. (2025). arXiv:2509.05291.

Dumas, T., et al. (2026). When Meanings Meet: Investigating the Emergence and Quality of Shared Concept Spaces during Multilingual Language Model Training. arXiv:2601.22851.

Hadad, I., et al. (2025). Formal Mechanistic Interpretability: Automated Circuit Discovery with Provable Guarantees. arXiv:2602.16823.

Jian, J., & Manning, C. D. (2026). Humans and transformer LMs: Abstraction drives language learning. EACL 2026.

Karhikeyan, K., Wang, Z., Mayhew, S., & Roth, D. (2020). Cross-lingual ability of multilingual BERT: An empirical study. ICLR 2020.

Kornblith, S., Norouzi, M., Lee, H., & Hinton, G. (2019). Similarity of neural network representations revisited. ICML 2019.

Lindsey, J., Gurnee, W., Ameisen, E., et al. (2025). On the Biology of a Large Language Model. Transformer Circuits Thread.

Liu, Z., Kitouni, O., Nolte, N., Michaud, E. J., Tegmark, M., & Williams, M. (2022). Towards understanding grokking: An effective theory of representation learning. NeurIPS 2022.

Loshchilov, I., & Hutter, F. (2019). Decoupled Weight Decay Regularization. ICLR 2019.

Nanda, N., Chan, L., Lieberum, T., Smith, J., & Steinhardt, J. (2023). Progress measures for grokking via mechanistic interpretability. ICLR 2023.

Power, A., Burda, Y., Edwards, H., Babuschkin, I., & Misra, V. (2022). Grokking: Generalization beyond overfitting on small algorithmic datasets. arXiv:2201.02177.

Syed, A., Rager, C., & Conmy, A. (2024). Attribution Patching Outperforms Automated Circuit Discovery. BlackboxNLP Workshop, EMNLP 2024.

Templeton, A., Conerly, T., Marcus, J., et al. (2024). Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. Anthropic.

Tracing Multilingual Representations in LLMs with Cross-Layer Transcoders. (2025). arXiv:2511.10840.

Wang, H., Minervini, P., & Ponti, E. (2024). Probing the Emergence of Cross-lingual Alignment during LLM Training. ACL Findings 2024.

Wendler, C., et al. (2025). Revisiting Multilingual Data Mixtures in Language Model Pretraining. arXiv:2510.25947.